DOCUMENT 02

Data Analysis

Data areas, preparation workflow, analytical questions, and data quality

Project: Olympic Sports Performance Analysis

Application: Python Flask web application | **Deployment:** Render

1. Purpose of Data Analysis

Data analysis is the process of examining structured information to identify patterns, compare groups, and communicate useful observations. In this project, the intended subject is Olympic sports performance.

2. Data Areas and Possible Variables

The project description indicates interest in medal achievements, athlete participation, sports performance, and country-level comparisons. Depending on the selected source dataset, useful fields may include athlete name or identifier, country/National Olympic Committee, sport, event, year, season, medal type, and participation details.

These are suggested fields for analysis, not a claim that every field is present in the current project dataset.

3. Recommended Analysis Workflow

Step 1 — Data collection

Select a reliable Olympic dataset and record its source, coverage, file format, and usage conditions.

Step 2 — Data understanding

Review column names, data types, row counts, missing values, duplicate records, and the meaning of each field.

Step 3 — Data cleaning

Standardise country and sport labels, handle missing or inconsistent values, check duplicates, and validate year and medal categories.

Step 4 — Exploratory analysis

Summarise participation and medal counts by year, country, sport, event, and medal category where the dataset supports those comparisons.

Step 5 — Interpretation

Look for broad trends and differences while avoiding unsupported claims about why a country or athlete performed in a particular way.

4. Example Questions for Analysis

- How does athlete participation vary across Olympic years?
- Which countries have the highest medal counts in the selected period?
- How are medals distributed across sports and medal categories?
- How has participation in selected sports changed over time?
- What differences appear between Summer and Winter Olympic data, if both are included?

5. Data Quality and Limitations

Historical Olympic datasets may use changing country names, event definitions, and competition formats. A medal record is not the same as an athlete count, and a country-level total may depend on how team events and historical national entities are represented.

Before publishing results, document the dataset source, date range, filtering rules, treatment of missing values, and any assumptions. Avoid comparing totals across periods without explaining differences in the number and type of events.

6. Outputs to Record

- A cleaned dataset or documented data-preparation workflow.
- Summary tables for participation and medal counts.
- Charts that answer clearly stated analytical questions.
- A short written interpretation for each major chart.

Documentation note: This document describes the project files supplied for review. The supplied ZIP does not include the source Olympic dataset or completed Tableau workbook. Therefore, no medal totals, rankings, chart values, or statistical findings are invented here. Update the relevant sections after connecting the final dataset and publishing the Tableau views.